

T _T Run ID	T _T Module	T _T Form	T _T Route
QA-sample-qa-test-project-20260805T1420	Customers	Customer edit	/staging/customers/{id}/edit
QA-sample-qa-test-project-20260805T1420	Customers	Customer edit	/staging/customers/{id}/edit
QA-sample-qa-test-project-20260805T1420	Customers	Customer edit	/staging/customers/{id}/edit
QA-sample-qa-test-project-20260805T1420	Customers	Customer edit	/staging/customers/{id}/edit
QA-sample-qa-test-project-20260805T1420	Dashboard	Dashboard inline Quick no	/staging
QA-sample-qa-test-project-20260805T1420	Dashboard	Dashboard inline Quick no	/staging
QA-sample-qa-test-project-20260805T1420	Identity	Registration	/staging/forms/registration
QA-sample-qa-test-project-20260805T1420	Identity	Registration	/staging/forms/registration
QA-sample-qa-test-project-20260805T1420	Identity	Registration	/staging/forms/registration
QA-sample-qa-test-project-20260805T1420	Identity	Registration	/staging/forms/registration
QA-sample-qa-test-project-20260805T1420	Orders	Order	/staging/forms/order
QA-sample-qa-test-project-20260805T1420	Orders	Order	/staging/forms/order
QA-sample-qa-test-project-20260805T1420	Orders	Order	/staging/forms/order
QA-sample-qa-test-project-20260805T1420	Orders	Order	/staging/forms/order
QA-sample-qa-test-project-20260805T1420	Orders	Order	/staging/forms/order
QA-sample-qa-test-project-20260805T1420	Orders	Order	/staging/forms/order
QA-sample-qa-test-project-20260805T1420	Orders	Order	/staging/forms/order
QA-sample-qa-test-project-20260805T1420	Orders	Order item	/staging/forms/order-item
QA-sample-qa-test-project-20260805T1420	Orders	Order item	/staging/forms/order-item
QA-sample-qa-test-project-20260805T1420	Orders	Order item	/staging/forms/order-item
QA-sample-qa-test-project-20260805T1420	Orders	Order item	/staging/forms/order-item

T _T Run ID	T _T Module	T _T Form	T _T Route
QA-sample-qa-test-project-20260805T1420	Orders	Order item	/staging/forms/order-item
QA-sample-qa-test-project-20260805T1420	Admin	QA Admin inventory	/staging/admin
QA-sample-qa-test-project-20260805T1420	Admin	QA Admin inventory	/staging/admin
QA-sample-qa-test-project-20260805T1420	Catalog	Product	/staging/forms/product
QA-sample-qa-test-project-20260805T1420	Catalog	Product	/staging/forms/product
QA-sample-qa-test-project-20260805T1420	Catalog	Product	/staging/forms/product
QA-sample-qa-test-project-20260805T1420	Catalog	Product	/staging/forms/product
QA-sample-qa-test-project-20260805T1420	Catalog	Product	/staging/forms/product
QA-sample-qa-test-project-20260805T1420	Catalog	Product	/staging/forms/product
QA-sample-qa-test-project-20260805T1420	Catalog	Product	/staging/forms/product
QA-sample-qa-test-project-20260805T1420	Catalog	Product	/staging/forms/product
QA-sample-qa-test-project-20260805T1420	Customers	Customer create	/staging/forms/customer-create
QA-sample-qa-test-project-20260805T1420	Customers	Customer create	/staging/forms/customer-create
QA-sample-qa-test-project-20260805T1420	Customers	Customer create	/staging/forms/customer-create
QA-sample-qa-test-project-20260805T1420	Customers	Customer create	/staging/forms/customer-create
QA-sample-qa-test-project-20260805T1420	Customers	Customer create	/staging/forms/customer-create
QA-sample-qa-test-project-20260805T1420	Customers	Customer create	/staging/forms/customer-create
QA-sample-qa-test-project-20260805T1420	Customers	Customer create	/staging/forms/customer-create
QA-sample-qa-test-project-20260805T1420	Customers	Customer create	/staging/forms/customer-create
QA-sample-qa-test-project-20260805T1420	Customers	Customer edit	/staging/customers/{id}/edit

T _T Run ID	T _T Module	T _T Form	T _T Route
QA-sample-qa-test-project-20260805T142C	Customers	Customer edit	/staging/customers/{id}/edit
QA-sample-qa-test-project-20260805T142C	Identity	Registration	/staging/forms/registration
QA-sample-qa-test-project-20260805T142C	Identity	Registration	/staging/forms/registration
QA-sample-qa-test-project-20260805T142C	Identity	Registration	/staging/forms/registration
QA-sample-qa-test-project-20260805T142C	Orders	Order	/staging/forms/order
QA-sample-qa-test-project-20260805T142C	Orders	Order	/staging/forms/order
QA-sample-qa-test-project-20260805T142C	Orders	Order	/staging/forms/order
QA-sample-qa-test-project-20260805T142C	Orders	Order	/staging/forms/order
QA-sample-qa-test-project-20260805T142C	Orders	Order	/staging/forms/order
QA-sample-qa-test-project-20260805T142C	Orders	Order item	/staging/forms/order-item
QA-sample-qa-test-project-20260805T142C	Orders	Order item	/staging/forms/order-item
QA-sample-qa-test-project-20260805T142C	Orders	Order item	/staging/forms/order-item
QA-sample-qa-test-project-20260805T142C	Admin	QA Admin inventory	/staging/admin
QA-sample-qa-test-project-20260805T142C	Admin	QA Admin inventory	/staging/admin
QA-sample-qa-test-project-20260805T142C	Admin	QA Admin inventory	/staging/admin
QA-sample-qa-test-project-20260805T142C	Admin	QA Admin inventory	/staging/admin
QA-sample-qa-test-project-20260805T142C	Authentication	Staging login	/staging/login
QA-sample-qa-test-project-20260805T142C	Authentication	Staging login	/staging/login
QA-sample-qa-test-project-20260805T142C	Catalog	Product	/staging/forms/product
QA-sample-qa-test-project-20260805T142C	Catalog	Product	/staging/forms/product
QA-sample-qa-test-project-20260805T142C	Catalog	Product	/staging/forms/product

T _T Run ID	T _T Module	T _T Form	T _T Route
QA-sample-qa-test-project-20260805T1420	Catalog	Product	/staging/forms/product
QA-sample-qa-test-project-20260805T1420	Catalog	Product	/staging/forms/product
QA-sample-qa-test-project-20260805T1420	Catalog	Product	/staging/forms/product
QA-sample-qa-test-project-20260805T1420	Catalog	Product	/staging/forms/product
QA-sample-qa-test-project-20260805T1420	Catalog	Product	/staging/forms/product
QA-sample-qa-test-project-20260805T1420	Catalog	Product	/staging/forms/product
QA-sample-qa-test-project-20260805T1420	Customers	Customer create	/staging/forms/customer-create
QA-sample-qa-test-project-20260805T1420	Customers	Customer create	/staging/forms/customer-create
QA-sample-qa-test-project-20260805T1420	Customers	Customer create	/staging/forms/customer-create
QA-sample-qa-test-project-20260805T1420	Customers	Customer create	/staging/forms/customer-create
QA-sample-qa-test-project-20260805T1420	Customers	Customer create	/staging/forms/customer-create
QA-sample-qa-test-project-20260805T1420	Customers	Customer create	/staging/forms/customer-create
QA-sample-qa-test-project-20260805T1420	Customers	Customer create	/staging/forms/customer-create
QA-sample-qa-test-project-20260805T1420	Customers	Customer create	/staging/forms/customer-create
QA-sample-qa-test-project-20260805T1420	Customers	Customer create	/staging/forms/customer-create
QA-sample-qa-test-project-20260805T1420	Customers	Customer edit	/staging/customers/{id}/edit
QA-sample-qa-test-project-20260805T1420	Customers	Customer edit	/staging/customers/{id}/edit
QA-sample-qa-test-project-20260805T1420	Dashboard	Dashboard inline Quick no	/staging
QA-sample-qa-test-project-20260805T1420	Dashboard	Dashboard inline Quick no	/staging
QA-sample-qa-test-project-20260805T1420	Identity	Registration	/staging/forms/registration

T _T Run ID	T _T Module	T _T Form	T _T Route
QA-sample-qa-test-project-20260805T142C	Identity	Registration	/staging/forms/registration
QA-sample-qa-test-project-20260805T142C	Identity	Registration	/staging/forms/registration
QA-sample-qa-test-project-20260805T142C	Identity	Registration	/staging/forms/registration
QA-sample-qa-test-project-20260805T142C	Identity	Registration	/staging/forms/registration
QA-sample-qa-test-project-20260805T142C	Orders	Order	/staging/forms/order
QA-sample-qa-test-project-20260805T142C	Orders	Order	/staging/forms/order
QA-sample-qa-test-project-20260805T142C	Orders	Order	/staging/forms/order
QA-sample-qa-test-project-20260805T142C	Orders	Order	/staging/forms/order
QA-sample-qa-test-project-20260805T142C	Orders	Order	/staging/forms/order
QA-sample-qa-test-project-20260805T142C	Orders	Order	/staging/forms/order
QA-sample-qa-test-project-20260805T142C	Orders	Order	/staging/forms/order
QA-sample-qa-test-project-20260805T142C	Orders	Order	/staging/forms/order
QA-sample-qa-test-project-20260805T142C	Orders	Order item	/staging/forms/order-item
QA-sample-qa-test-project-20260805T142C	Orders	Order item	/staging/forms/order-item
QA-sample-qa-test-project-20260805T142C	Orders	Order item	/staging/forms/order-item
QA-sample-qa-test-project-20260805T142C	Orders	Order item	/staging/forms/order-item
QA-sample-qa-test-project-20260805T142C	Orders	Order item	/staging/forms/order-item
QA-sample-qa-test-project-20260805T142C	Orders	Order item	/staging/forms/order-item
QA-sample-qa-test-project-20260805T142C	Orders	Order item	/staging/forms/order-item
QA-sample-qa-test-project-20260805T142C	Orders	Order item	/staging/forms/order-item
QA-sample-qa-test-project-20260805T142C	System	System-managed entity ti	No editable date form

T _T Run ID	T _T Module	T _T Form	T _T Route
QA-sample-qa-test-project-20260805T1420	Admin	QA Admin inventory	/staging/admin
QA-sample-qa-test-project-20260805T1420	Catalog	Product	/staging/forms/product
QA-sample-qa-test-project-20260805T1420	Catalog	Product	/staging/forms/product
QA-sample-qa-test-project-20260805T1420	Customers	Customer create	/staging/forms/customer-create
QA-sample-qa-test-project-20260805T1420	Customers	Customer edit	/staging/customers/{id}/edit
QA-sample-qa-test-project-20260805T1420	Customers	Customer edit	/staging/customers/{id}/edit
QA-sample-qa-test-project-20260805T1420	Dashboard	Dashboard inline Quick navigation	/staging
QA-sample-qa-test-project-20260805T1420	Identity	Registration	/staging/forms/registration
QA-sample-qa-test-project-20260805T1420	Identity	Registration	/staging/forms/registration
QA-sample-qa-test-project-20260805T1420	Orders	Order	/staging/forms/order
QA-sample-qa-test-project-20260805T1420	Orders	Order	/staging/forms/order
QA-sample-qa-test-project-20260805T1420	Orders	Order item	/staging/forms/order-item
QA-sample-qa-test-project-20260805T1420	Orders	Order item	/staging/forms/order-item
QA-sample-qa-test-project-20260805T1420	Admin	QA Admin inventory	/staging/admin
QA-sample-qa-test-project-20260805T1420	Catalog	Product	/staging/forms/product
QA-sample-qa-test-project-20260805T1420	Catalog	Product	/staging/forms/product
QA-sample-qa-test-project-20260805T1420	Customers	Customer create	/staging/forms/customer-create
QA-sample-qa-test-project-20260805T1420	Customers	Customer create	/staging/forms/customer-create
QA-sample-qa-test-project-20260805T1420	Customers	Customer create	/staging/forms/customer-create
QA-sample-qa-test-project-20260805T1420	Customers	Customer create	/staging/forms/customer-create
QA-sample-qa-test-project-20260805T1420	Identity	Registration	/staging/forms/registration

T _T Run ID	T _T Module	T _T Form	T _T Route
QA-sample-qa-test-project-20260805T142C	Orders	Order	/staging/forms/order
QA-sample-qa-test-project-20260805T142C	Orders	Order	/staging/forms/order
QA-sample-qa-test-project-20260805T142C	Orders	Order	/staging/forms/order
QA-sample-qa-test-project-20260805T142C	Orders	Order	/staging/forms/order

T _T Field	T _T Type / required	T _T Test scenario
delta	integer; required; final stock >=0	Schema-valid domain-specific delta
product_id	integer FK; required	Schema-valid domain-specific produ
reason	string; optional	Schema-valid domain-specific reasc
password	string; required	Schema-valid domain-specific passv
username	string; required	Schema-valid domain-specific userr
active	boolean/integer; default true	Schema-valid domain-specific activ
category	enum; required	Schema-valid domain-specific categ
description	string; optional; <=1000	Schema-valid domain-specific descri
name	string; required; 2-120	Schema-valid domain-specific name
sku	string; required; unique; 3-24	Schema-valid domain-specific sku
stock_quantity	integer; default 0; 0..1000000	Schema-valid domain-specific stock
unit_price	decimal(10,2); required; >0	Schema-valid domain-specific unit_p
address_line1	string; required; 3-120	Schema-valid domain-specific addre
address_line2	string; optional; <=120	Schema-valid domain-specific addre
city	string; required; 2-80	Schema-valid domain-specific city
email	email string; required; unique	Schema-valid domain-specific emai
full_name	string; required; 2-100	Schema-valid domain-specific full_n
notes	string; optional; <=1000	Schema-valid domain-specific notes
phone	string; optional; 7-24 phone char	Schema-valid domain-specific phon
postal_code	string; required; 3-20	Schema-valid domain-specific posta
status	enum; required/default	Schema-valid domain-specific statu

T _T Field	T _T Type / required	T _T Test scenario
email	email string; required; unique	Schema-valid domain-specific email
full_name	string; required; 2-100	Schema-valid domain-specific full_name
notes	string; optional	Schema-valid domain-specific notes
phone	string; optional; 7-24 phone char	Schema-valid domain-specific phone
note	string; required	Schema-valid domain-specific note
replay_token	string; required hidden	Schema-valid domain-specific replay_token
display_name	string; optional; <=80	Schema-valid domain-specific display_name
email	email string; required; unique	Schema-valid domain-specific email
role	enum; required/default	Schema-valid domain-specific role
username	string; required; unique; 3-40	Schema-valid domain-specific username
customer_id	integer FK; required	Schema-valid domain-specific customer_id
gift_message	string; optional; <=240	Schema-valid domain-specific gift_message
product_id	integer FK; required	Schema-valid domain-specific product_id
quantity	integer; required; 1..100	Schema-valid domain-specific quantity
shipping_address	string; required; 1-200	Schema-valid domain-specific shipping_address
shipping_method	enum; default standard	Schema-valid domain-specific shipping_method
status	enum; default draft	Schema-valid domain-specific status
order_id	integer FK; required	Schema-valid domain-specific order_id
order_id	integer FK; required	Run-tagged existing order FK
product_id	integer FK; required	Schema-valid domain-specific product_id
quantity	integer; required; positive; <= stock	Schema-valid domain-specific quantity

T _T Field	T _T Type / required	T _T Test scenario
unit_price	decimal(10,2); required; >=0	Schema-valid domain-specific unit_
delta	integer; required; final stock >=0	Maximum safe decrement to zero
reason	string; optional	Optional field omitted / empty
active	boolean/integer; default true	Valid checkbox false
description	string; optional; <=1000	Optional field omitted / empty
sku	string; required; unique; 3-24	Minimum valid SKU length 3
sku	string; required; unique; 3-24	Maximum valid SKU length 24
stock_quantity	integer; default 0; 0..1000000	Minimum nonnegative stock
stock_quantity	integer; default 0; 0..1000000	Maximum allowed stock
unit_price	decimal(10,2); required; >0	Minimum valid positive two-decimal
unit_price	decimal(10,2); required; >0	Maximum valid amount
address_line1	string; required; 3-120	Minimum valid address length=3
address_line2	string; optional; <=120	Optional field omitted / empty
city	string; required; 2-80	Minimum valid city length=2
full_name	string; required; 2-100	Exactly minimum length 2
notes	string; optional; <=1000	Optional field omitted / empty
notes	string; optional; <=1000	Exactly 1000 characters
phone	string; optional; 7-24 phone char	Optional field omitted / empty
phone	string; optional; 7-24 phone char	Valid minimum seven phone digits
postal_code	string; required; 3-20	Minimum valid postal length=3
notes	string; optional	Optional field omitted / empty

T _T Field	T _T Type / required	T _T Test scenario
phone	string; optional; 7-24 phone char	Optional field omitted / empty
display_name	string; optional; <=80	Optional field omitted / empty
username	string; required; unique; 3-40	Minimum valid username length=3
username	string; required; unique; 3-40	Maximum valid username length=40
gift_message	string; optional; <=240	Optional field omitted / empty
gift_message	string; optional; <=240	Maximum valid message length 240
quantity	integer; required; 1..100	Minimum order quantity
quantity	integer; required; 1..100	Maximum schema quantity
shipping_address	string; required; 1-200	Exactly 200-character address
quantity	integer; required; positive; <= stock	Minimum valid item quantity
quantity	integer; required; positive; <= stock	Maximum valid quantity equal to stock
unit_price	decimal(10,2); required; >=0	Minimum allowed zero
delta	integer; required; final stock >=0	Missing mandatory field
delta	integer; required; final stock >=0	Unsafe decrement beyond stock
product_id	integer FK; required	Missing mandatory field
product_id	integer FK; required	Unknown inventory product
password	string; required	Missing mandatory field
username	string; required	Missing mandatory field
category	enum; required	Missing mandatory field
category	enum; required	Disallowed product category
description	string; optional; <=1000	Inert reflected markup injection

T _T Field	T _T Type / required	T _T Test scenario
name	string; required; 2-120	Missing mandatory field
sku	string; required; unique; 3-24	Missing mandatory field
sku	string; required; unique; 3-24	Invalid punctuation
stock_quantity	integer; default 0; 0..1000000	Negative stock
unit_price	decimal(10,2); required; >0	Missing mandatory field
unit_price	decimal(10,2); required; >0	Zero price against ORM validator
address_line1	string; required; 3-120	Missing mandatory field
city	string; required; 2-80	Missing mandatory field
email	email string; required; unique	Missing mandatory field
email	email string; required; unique	Malformed customer email
full_name	string; required; 2-100	Missing mandatory field
notes	string; optional; <=1000	Exceeds maximum by one character
phone	string; optional; 7-24 phone char	Invalid non-phone characters
postal_code	string; required; 3-20	Missing mandatory field
status	enum; required/default	Missing mandatory field
status	enum; required/default	Disallowed customer status
email	email string; required; unique	Missing mandatory field
full_name	string; required; 2-100	Missing mandatory field
note	string; required	Missing mandatory field
replay_token	string; required hidden	Missing mandatory field
email	email string; required; unique	Missing mandatory field

T _T Field	T _T Type / required	T _T Test scenario
role	enum; required/default	Missing mandatory field
role	enum; required/default	Disallowed role
username	string; required; unique; 3-40	Missing mandatory field
username	string; required; unique; 3-40	Above maximum username length
customer_id	integer FK; required	Missing mandatory field
customer_id	integer FK; required	Nonexistent customer reference
product_id	integer FK; required	Missing mandatory field
product_id	integer FK; required	Nonexistent product reference
quantity	integer; required; 1..100	Missing mandatory field
quantity	integer; required; 1..100	Zero quantity
shipping_address	string; required; 1-200	Missing mandatory field
status	enum; default draft	Disallowed order status
order_id	integer FK; required	Missing mandatory field
order_id	integer FK; required	Nonexistent order reference
product_id	integer FK; required	Missing mandatory field
product_id	integer FK; required	Nonexistent product reference
quantity	integer; required; positive; <= sto	Missing mandatory field
quantity	integer; required; positive; <= sto	One above available stock
unit_price	decimal(10,2); required; >=0	Missing mandatory field
unit_price	decimal(10,2); required; >=0	Negative unit price
created_at / last_seeded	datetime; generated server-side	Past/future date applicability review

T _T Field	T _T Type / required	T _T Test scenario
delta	integer; required; final stock >=0	Safe zero-change update
sku	string; required; unique; 3-24	Duplicate unique product SKU
unit_price	decimal(10,2); required; >0	Repeated negative ORM price
email	email string; required; unique	Duplicate unique customer email
email	email string; required; unique	Same malformed email, different error
phone	string; optional; 7-24 phone char	Repeat same invalid phone in create
note	string; required	Transient first response must pass i
email	email string; required; unique	Malformed email error comparison
username	string; required; unique; 3-40	Duplicate unique username
quantity	integer; required; 1..100	Quantity exceeds available stock
shipping_address	string; required; 1-200	Required empty shipping address
product_id	integer FK; required	Duplicate composite order/product
quantity	integer; required; positive; <= sto	Repeated negative quantity
reason	string; optional	Unicode explanatory reason
category	enum; required	Alternate allowed categories
description	string; optional; <=1000	Inert Unicode / special characters
address_line1	string; required; 3-120	Unicode fictional address
city	string; required; 2-80	Unicode city
full_name	string; required; 2-100	Unicode and accents
status	enum; required/default	Alternate allowed suspended status
role	enum; required/default	Alternate allowed role

T _T Field	T _T Type / required	T _T Test scenario
customer_id	integer FK; required	Existing Unicode customer FK
product_id	integer FK; required	Existing alternate product FK
shipping_method	enum; default standard	Alternate valid pickup method
status	enum; default draft	Safe draft status

T _T Value generated	T _T Expected output
0	Accept valid field and preserve business rules
4	Accept valid field and preserve business rules
QA-sample-qa-test-project-20260805T142052 zero-c	Accept valid field and preserve business rules
SAMPLE-QA-PASSWORD-NOT-SECRET	Accept valid field and preserve business rules
qa.staging.runner	Accept valid field and preserve business rules
true	Accept valid field and preserve business rules
test-kit	Accept valid field and preserve business rules
QA-sample-qa-test-project-20260805T142052 reusal	Accept valid field and preserve business rules
QA-sample-qa-test-project-20260805T142052 Samp	Accept valid field and preserve business rules
QA-142052-VALID	Accept valid field and preserve business rules
8	Accept valid field and preserve business rules
24.95	Accept valid field and preserve business rules
44 Sandbox Commerce Lane	Accept valid field and preserve business rules
Suite QA-4	Accept valid field and preserve business rules
Example Springs	Accept valid field and preserve business rules
qa.20260805t142052.customer@example.com	Accept valid field and preserve business rules
QA-sample-qa-test-project-20260805T142052 Taylor	Accept valid field and preserve business rules
QA-sample-qa-test-project-20260805T142052 reusal	Accept valid field and preserve business rules
+1-555-0199	Accept valid field and preserve business rules
142052	Accept valid field and preserve business rules
active	Accept valid field and preserve business rules

T _T Value generated	T _T Expected output
qa.20260805t142052.customer@example.com	Accept valid field and preserve business rules
QA-sample-qa-test-project-20260805T142052 Taylor	Accept valid field and preserve business rules
QA-sample-qa-test-project-20260805T142052 verifie	Accept valid field and preserve business rules
+1-555-0198	Accept valid field and preserve business rules
QA-sample-qa-test-project-20260805T142052 deterr	Accept valid field and preserve business rules
false-positive-stable-token	Accept valid field and preserve business rules
QA-sample-qa-test-project-20260805T142052	Accept valid field and preserve business rules
qa.20260805t142052.registration@example.com	Accept valid field and preserve business rules
qa_editor	Accept valid field and preserve business rules
qa_20260805_142052	Accept valid field and preserve business rules
4	Accept valid field and preserve business rules
QA-sample-qa-test-project-20260805T142052 positi	Accept valid field and preserve business rules
4	Accept valid field and preserve business rules
2	Accept valid field and preserve business rules
44 Sandbox Commerce Lane, Example Springs 1420	Accept valid field and preserve business rules
standard	Accept valid field and preserve business rules
draft	Accept valid field and preserve business rules
2	Accept valid field and preserve business rules
2	Accept existing orders.id=2
2	Accept valid field and preserve business rules
1	Accept valid field and preserve business rules

T _T Value generated	T _T Expected output
42.50	Accept valid field and preserve business rules
-8	Accept only when generated product current stock
	Accept omitted optional field and apply documente
false	Accept inactive product
	Accept omitted optional field and apply documente
QA1	Accept minimum 3 characters
QA-142052-BOUNDARY-ABCDE	Accept exactly 24 uppercase/digit/hyphen charact
0	Accept stock_quantity=0
1000000	Accept max stock boundary
0.01	Accept positive amount
999999.99	Accept schema maximum with 2 decimal places
7 QA	Accept exactly 3 characters
	Accept omitted optional field and apply documente
Xi	Accept exactly 2 characters
Qi	Accept exactly two characters
	Accept omitted optional field and apply documente
QA-sample-qa-test-project-20260805T142052 ñññññ	Accept max-length notes
	Accept omitted optional field and apply documente
5550101	Accept exactly seven phone characters
QA1	Accept exactly 3 characters
	Accept omitted optional field and apply documente

T _T Value generated	T _T Expected output
	Accept omitted optional field and apply documente
	Accept omitted optional field and apply documente
Qa1	Accept exactly 3 characters
Qaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa	Accept exactly 40 characters
	Accept omitted optional field and apply documente
QA-sample-qa-test-project-20260805T142052 Fictio	Accept 240 characters
1	Accept quantity=1 when stock permits
100	Accept only when product stock >=100
QA-sample-qa-test-project-20260805T142052 55 Fic	Accept 200 characters
1	Accept quantity=1 when stock permits
4	Accept quantity=4 for product id=2
0.00	Accept nonnegative schema boundary
	Reject missing required input before persistence
-999999	Reject negative resulting stock
	Reject missing required input before persistence
999999	Reject missing product with 404
	Reject missing required input before persistence
	Reject missing required input before persistence
	Reject missing required input before persistence
pharmaceutical	Reject category outside declared enum
<b data-qa-oracle='GAP-04'>SAMPLE TEST MARKUP	Escape markup as visible text

T _T Value generated	T _T Expected output
	Reject missing required input before persistence
	Reject missing required input before persistence
AB!	Reject invalid characters
-1	Reject stock below zero
	Reject missing required input before persistence
0.00	Reject non-positive ORM value
	Reject missing required input before persistence
	Reject missing required input before persistence
	Reject missing required input before persistence
not-an-email	Reject invalid format
	Reject missing required input before persistence
QA-sample-qa-test-project-20260805T142052 ñññññ	Reject 1001-character notes
NOT-A-PHONE	Reject; expected ^\+?[0-9 ()-]{7,24}\$
	Reject missing required input before persistence
	Reject missing required input before persistence
deleted	Reject value outside active inactive suspended
	Reject missing required input before persistence
	Reject missing required input before persistence
	Reject missing required input before persistence
	Reject missing required input before persistence
	Reject missing required input before persistence

T _T Value generated	T _T Expected output
	Reject missing required input before persistence
billing_admin	Reject value outside qa_viewer qa_editor
	Reject missing required input before persistence
Qaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa	Reject 41-character username
	Reject missing required input before persistence
999999	Reject before order creation
	Reject missing required input before persistence
999999	Reject before order creation
	Reject missing required input before persistence
0	Reject before persistence
	Reject missing required input before persistence
refunded	Reject value outside declared enum
	Reject missing required input before persistence
999999	Reject missing orders.id
	Reject missing required input before persistence
999999	Reject missing products.id
	Reject missing required input before persistence
5	Reject quantity=5 for product stock=4
	Reject missing required input before persistence
-0.01	Reject database CHECK violation
2025-01-01 / 2027-01-01	Never invent or submit a user date when no editabl

T† Value generated	T† Expected output
0	Preserve existing stock while exercising admin han
QA-142052-VALID	Reject duplicate SKU
-3.25	Reject unit_price <= 0
qa.20260805t142052.customer@example.com	Reject duplicate email
bad	Reject with same wording as registration
ALSO-BAD	Reject phone outside ^\+?[0-9 ()-]{7,24}\$
QA-sample-qa-test-project-20260805T142052 deterr	Second identical submission succeeds; do not log
bad	Reject with consistent customer-edit message
qa_20260805_142052	Reject duplicate username
5	Reject quantity=5 for product stock=4
	Reject empty required address
2 / 4	Reject duplicate (order_id, product_id)
-7	Reject quantities <=0
QA-sample-qa-test-project-20260805T142052 café c	Accept optional Unicode reason
office electronics home test-kit	Accept enum members only
Café kit — 日本語 / & <quoted>	Escape display while preserving Unicode
77 Café Sandbox Straße	Accept valid Unicode address
Málaga	Accept Unicode city
QA-sample-qa-test-project-20260805T142052 Zoë N	Accept valid Unicode name within 100 characters
suspended	Accept enum member
qa_viewer	Accept allowed alternate role

T _T	Value generated	T _T	Expected output
6			Accept existing customers.id=6
2			Accept existing products.id=2
pickup			Accept declared enum member
draft			Accept and trigger no external actions

[illegible]

T _T Output / actual	T _T Validation type	🗂 Dataset category
Accepted during live positive/boundary workflow	client + server + schema	happy path
Accepted during live positive/boundary workflow	client + server + schema	happy path
Accepted during live positive/boundary workflow	client + server + schema	happy path
Accepted during live positive/boundary workflow	client + server + schema	happy path
First transient failure; identical retry succeeded (not a g	client + server + schema	happy path
First transient failure; identical retry succeeded (not a g	client + server + schema	happy path
Accepted during live positive/boundary workflow	client + server + schema	happy path
Accepted during live positive/boundary workflow	client + server + schema	happy path
Accepted during live positive/boundary workflow	client + server + schema	happy path
Accepted during live positive/boundary workflow	client + server + schema	happy path
Accepted during live positive/boundary workflow	client + server + schema	happy path
Accepted during live positive/boundary workflow	client + server + schema	happy path
Accepted during live positive/boundary workflow	client + server + schema	happy path
Accepted during live positive/boundary workflow	client + server + schema	happy path
Accepted during live positive/boundary workflow	client + server + schema	happy path
Accepted during live positive/boundary workflow	client + server + schema	happy path
Accepted; parent gift message contains run identifier	foreign key relationship	happy path
Accepted during live positive/boundary workflow	client + server + schema	happy path
Accepted during live positive/boundary workflow	client + server + schema	happy path

T _T Output / actual	T _T Validation type	Dataset category
Accepted during live positive/boundary workflow	client + server + schema	happy path
Generated; final stock would be zero	conditional minimum stock	boundary
Generated valid omission; submit only within a complete	optionality / defaults	boundary
Generated from bool/default	boolean conditional	boundary
Generated valid omission; submit only within a complete	optionality / defaults	boundary
Generated from SKU_PATTERN	min boundary	boundary
Generated from max_length=24	max boundary	boundary
Schema and DB permit zero	min boundary / DB CHECK	boundary
Generated from ProductCreate le=1000000	max boundary	boundary
Generated from Product.validate_unit_price > 0	min boundary	boundary
Generated from ProductCreate le=999999.99	max boundary	boundary
Generated from CustomerCreate min_length=3	min boundary	boundary
Generated valid omission; submit only within a complete	optionality / defaults	boundary
Generated from CustomerCreate min_length=2	min boundary	boundary
Generated; keep run tag in notes if executed	min boundary	boundary
Generated valid omission; submit only within a complete	optionality / defaults	boundary
Accepted; Unicode notes length=1000	max boundary / unicode	boundary
Generated valid omission; submit only within a complete	optionality / defaults	boundary
Generated from PHONE_PATTERN	min boundary / format	boundary
Generated from min_length=3	min boundary	boundary
Generated valid omission; submit only within a complete	optionality / defaults	boundary

T _T Output / actual	T _T Validation type	Dataset category
Generated valid omission; submit only within a complete	optionality / defaults	boundary
Generated valid omission; submit only within a complete	optionality / defaults	boundary
Accepted and created tagged registration	min boundary	boundary
Accepted and created tagged registration	max boundary	boundary
Generated valid omission; submit only within a complete	optionality / defaults	boundary
Generated from max_length=240	max boundary / run tagging	boundary
Accepted in empty-address replay; relation remained valid	min boundary / relationship	boundary
Seed product TEST-PAD-03 has stock=100; generated valid	max boundary / conditional stock	boundary
Generated from max_length=200	max boundary	boundary
Accepted for existing order=2 and product=2	min boundary / relationships	boundary
Generated from Product.stock_quantity=4	max boundary / conditional stock	boundary
Generated from ck_order_item_nonnegative_price	DB check minimum	boundary
Generated negative; not submitted individually	required field	negative
Server: adjustment would make stock negative	server-side business rule	negative
Generated negative; not submitted individually	required field	negative
Handler resolves product before any update	foreign key / safe lookup	negative
Generated negative; not submitted individually	required field	negative
Generated negative; not submitted individually	required field	negative
Generated negative; not submitted individually	required field	negative
Generated; never force an invalid select value	enum allowed list	negative
FAILED: markup became a live HTML element twice	stored/reflected output encoding	negative

T _T Output / actual	T _T Validation type	Dataset category
Generated negative; not submitted individually	required field	negative
Generated negative; not submitted individually	required field	negative
Server: SKU must use uppercase letters, numbers, and	format / server-side	negative
Browser numeric min and DB CHECK reject negatives	min-1 / DB CHECK	negative
Generated negative; not submitted individually	required field	negative
Generated; same rule as confirmed negative-price findi	ORM @validates	negative
Generated negative; not submitted individually	required field	negative
Generated negative; not submitted individually	required field	negative
Generated negative; not submitted individually	required field	negative
Generated; HTML email type blocks browser submissic	email format	negative
Generated negative; not submitted individually	required field	negative
Generated negative; browser maxlength caps typing	max+1 / excessive text	negative
FAILED: invalid phone was saved on a run-tagged custc	format / API schema	negative
Generated negative; not submitted individually	required field	negative
Generated negative; not submitted individually	required field	negative
Generated; select does not expose invalid value	enum allowed list	negative
Generated negative; not submitted individually	required field	negative
Generated negative; not submitted individually	required field	negative
Generated negative; not submitted individually	required field	negative
Generated negative; not submitted individually	required field	negative
Generated negative; not submitted individually	required field	negative

T _T Output / actual	T _T Validation type	Dataset category
Generated negative; not submitted individually	required field	negative
Generated; browser select exposes only allowed option	enum allowed list	negative
Chrome blocked submission: Please fill out this field.	required field	negative
Generated; not executed	max+1 / length	negative
Generated negative; not submitted individually	required field	negative
Handler checks related customer; browser select cannot	foreign key integrity	negative
Generated negative; not submitted individually	required field	negative
Handler resolves selected product before insert	foreign key integrity	negative
Generated negative; not submitted individually	required field	negative
Chrome: Value must be greater than or equal to 1.	client numeric min	negative
FAILED: accepted twice; see GAP-01 / TCW-89	required field	negative
Generated; browser select offers only declared values	enum allowed list	negative
Generated negative; not submitted individually	required field	negative
Would create an orphan because SQLite FK enforcement	foreign key integrity	negative
Generated negative; not submitted individually	required field	negative
Code oracle predicts persistence, which would create a	foreign key integrity / GAP-03 can	negative
Generated negative; not submitted individually	required field	negative
Order create path confirmed the same missing stock rule	application business rule	negative
Generated negative; not submitted individually	required field	negative
Generated; DB constraint prevents persistence	DB CHECK	negative
No in-scope form exposes an editable date; marked no	date field applicability	negative

T _T Output / actual	T _T Validation type	Dataset category
Accepted; existing stock remains 8	safe non-destructive admin	regression
Server: database constraint rejected product	database uniqueness	regression
FAILED: negative price persisted twice	ORM schema-level rule	regression
Server: email must be unique	uniqueness	regression
Customer edit: Enter a valid customer contact email.	format / message consistency	regression
FAILED: exact same edit accepted twice and persisted	format / schema	regression
First error followed by success; CONTROL-01 discarded	retest / false-positive control	regression
Registration: value is not a valid email address: An ema	format / consistency	regression
Server: username and email must be unique	uniqueness	regression
FAILED: two tagged orders and related items were crea	application business rule	regression
FAILED: two run-tagged orders were created with empty	required / ORM schema	regression
DDL declares UNIQUE(order_id,product_id)	composite uniqueness	regression
FAILED: two distinct valid relation pairs accepted -7	application business rule	regression
Generated valid Unicode reason	unicode / optional	UAT
test-kit accepted; remaining allowed values generated	enum allowed list	UAT
Generated safe positive text; no scripts/external links	unicode / output escaping	UAT
Accepted for run-tagged Unicode customer	unicode	UAT
Accepted for tagged Unicode customer	unicode	UAT
Accepted: Zoë Núñez Sample	unicode / max length	UAT
Generated; no external integration exists	enum allowed list	UAT
Accepted in minimum/maximum username registration	enum allowed list	UAT

T _T Output / actual	T _T Validation type	🔍 Dataset category
Accepted; over-stock probes retained a valid customer	foreign key relationship	UAT
Accepted; stock is 4	foreign key relationship	UAT
Generated; no fulfillment integration exists	enum allowed list	UAT
Accepted in all submitted orders	enum / external-side-effect safety	UAT

T _T Dataset class	T _T Rule / source at pinned commit
Reusable valid	app/main.py:255-261 @ 84d392ec8f0d30da10fccdbf00ea2bc
Reusable valid	app/main.py:255-261 @ 84d392ec8f0d30da10fccdbf00ea2bc
Reusable valid	app/main.py:255-261 @ 84d392ec8f0d30da10fccdbf00ea2bc
Reusable valid	README.md:33-40; app/main.py:106-117 @ 84d392ec8f0d30
Reusable valid	README.md:33-40; app/main.py:106-117 @ 84d392ec8f0d30
Reusable valid	app/models.py:58; app/schemas.py:40 @ 84d392ec8f0d30da
Reusable valid	app/models.py:13-18; app/schemas.py:37 @ 84d392ec8f0d30
Reusable valid	app/schemas.py:36; templates/index.html:18 @ 84d392ec8f0
Reusable valid	app/schemas.py:35; app/models.py:54 @ 84d392ec8f0d30da
Reusable valid	app/schemas.py:7,34,42-47; app/models.py:53 @ 84d392ec8f
Reusable valid	app/models.py:57,62; app/schemas.py:39 @ 84d392ec8f0d30
Reusable valid	app/models.py:56,64-74; app/schemas.py:38 @ 84d392ec8f0
Reusable valid	app/schemas.py:19; app/models.py:31 @ 84d392ec8f0d30da
Reusable valid	app/schemas.py:20; app/models.py:32 @ 84d392ec8f0d30da
Reusable valid	app/schemas.py:21; app/models.py:33 @ 84d392ec8f0d30da
Reusable valid	app/schemas.py:17; app/models.py:29 @ 84d392ec8f0d30da
Reusable valid	app/schemas.py:16; app/models.py:28 @ 84d392ec8f0d30da
Reusable valid	app/schemas.py:23; app/models.py:35 @ 84d392ec8f0d30da
Reusable valid	app/schemas.py:6,18,26-31 @ 84d392ec8f0d30da10fccdbf00
Reusable valid	app/schemas.py:22; app/models.py:34 @ 84d392ec8f0d30da
Reusable valid	app/models.py:7-11,36; app/schemas.py:24 @ 84d392ec8f0d

T _T Dataset class	T _T Rule / source at pinned commit
Reusable valid	app/models.py:29; app/main.py:182-194 @ 84d392ec8f0d30d
Reusable valid	app/models.py:28; app/main.py:182-194 @ 84d392ec8f0d30d
Reusable valid	app/models.py:35; app/main.py:182-194 @ 84d392ec8f0d30d
Reusable valid	app/schemas.py:6,26-31; app/main.py:182-194 @ 84d392ec8f0d30d
Reusable valid	app/main.py:245-251; templates/index.html:17 @ 84d392ec8f0d30d
Reusable valid	app/main.py:245-251; templates/index.html:17 @ 84d392ec8f0d30d
Reusable valid	app/schemas.py:12; app/models.py:45 @ 84d392ec8f0d30da
Reusable valid	app/schemas.py:10-11; app/models.py:44 @ 84d392ec8f0d30da
Reusable valid	app/schemas.py:13 @ 84d392ec8f0d30da10fccdbf00ea2bcfa
Reusable valid	app/schemas.py:9-13; app/models.py:40-48 @ 84d392ec8f0d30da
Reusable valid	app/models.py:80; app/schemas.py:51 @ 84d392ec8f0d30da
Reusable valid	app/schemas.py:56; app/models.py:84 @ 84d392ec8f0d30da
Reusable valid	app/models.py:92-93; app/schemas.py:52 @ 84d392ec8f0d30da
Reusable valid	app/schemas.py:53; README.md:82 @ 84d392ec8f0d30da10fccdbf00ea2bcfa
Reusable valid	app/schemas.py:57; app/models.py:81 @ 84d392ec8f0d30da
Reusable valid	app/schemas.py:55; app/models.py:83 @ 84d392ec8f0d30da
Reusable valid	app/models.py:20-24,82; app/schemas.py:54 @ 84d392ec8f0d30da
Reusable valid	app/models.py:92,99 @ 84d392ec8f0d30da10fccdbf00ea2bcfa
Reusable valid	app/models.py:92,99 @ 84d392ec8f0d30da10fccdbf00ea2bcfa
Reusable valid	app/models.py:93,99 @ 84d392ec8f0d30da10fccdbf00ea2bcfa
Reusable valid	README.md:82; app/main.py:233-242 @ 84d392ec8f0d30da10fccdbf00ea2bcfa

T _T Dataset class	T _T Rule / source at pinned commit
Reusable valid	app/models.py:94,100; migrations/versions/001_initial.sql @ 84d392ec8f0d30da10fccdbf00ea2bc
Reusable valid	app/main.py:255-261 @ 84d392ec8f0d30da10fccdbf00ea2bc
Reusable valid	app/main.py:255-261 @ 84d392ec8f0d30da10fccdbf00ea2bc
Reusable valid	app/models.py:58; app/schemas.py:40 @ 84d392ec8f0d30da10fccdbf00ea2bc
Reusable valid	app/schemas.py:36; templates/index.html:18 @ 84d392ec8f0d30da10fccdbf00ea2bc
Reusable valid	app/schemas.py:7,34,42-47; app/models.py:53 @ 84d392ec8f0d30da10fccdbf00ea2bc
Reusable valid	app/schemas.py:7,34,42-47; app/models.py:53 @ 84d392ec8f0d30da10fccdbf00ea2bc
Reusable valid	app/models.py:57,62; app/schemas.py:39 @ 84d392ec8f0d30da10fccdbf00ea2bc
Reusable valid	app/models.py:57,62; app/schemas.py:39 @ 84d392ec8f0d30da10fccdbf00ea2bc
Reusable valid	app/models.py:56,64-74; app/schemas.py:38 @ 84d392ec8f0d30da10fccdbf00ea2bc
Reusable valid	app/models.py:56,64-74; app/schemas.py:38 @ 84d392ec8f0d30da10fccdbf00ea2bc
Reusable valid	app/schemas.py:19; app/models.py:31 @ 84d392ec8f0d30da10fccdbf00ea2bc
Reusable valid	app/schemas.py:20; app/models.py:32 @ 84d392ec8f0d30da10fccdbf00ea2bc
Reusable valid	app/schemas.py:21; app/models.py:33 @ 84d392ec8f0d30da10fccdbf00ea2bc
Reusable valid	app/schemas.py:16; app/models.py:28 @ 84d392ec8f0d30da10fccdbf00ea2bc
Reusable valid	app/schemas.py:23; app/models.py:35 @ 84d392ec8f0d30da10fccdbf00ea2bc
Reusable valid	app/schemas.py:23; app/models.py:35 @ 84d392ec8f0d30da10fccdbf00ea2bc
Reusable valid	app/schemas.py:6,18,26-31 @ 84d392ec8f0d30da10fccdbf00ea2bc
Reusable valid	app/schemas.py:6,18,26-31 @ 84d392ec8f0d30da10fccdbf00ea2bc
Reusable valid	app/schemas.py:22; app/models.py:34 @ 84d392ec8f0d30da10fccdbf00ea2bc
Reusable valid	app/models.py:35; app/main.py:182-194 @ 84d392ec8f0d30da10fccdbf00ea2bc

T _T Dataset class	T _T Rule / source at pinned commit
Reusable valid	app/schemas.py:6,26-31; app/main.py:182-194 @ 84d392ec8
Reusable valid	app/schemas.py:12; app/models.py:45 @ 84d392ec8f0d30da
Reusable valid	app/schemas.py:9-13; app/models.py:40-48 @ 84d392ec8f0d
Reusable valid	app/schemas.py:9-13; app/models.py:40-48 @ 84d392ec8f0d
Reusable valid	app/schemas.py:56; app/models.py:84 @ 84d392ec8f0d30da
Reusable valid	app/schemas.py:56; app/models.py:84 @ 84d392ec8f0d30da
Reusable valid	app/schemas.py:53; README.md:82 @ 84d392ec8f0d30da10
Reusable valid	app/schemas.py:53; README.md:82 @ 84d392ec8f0d30da10
Reusable valid	app/schemas.py:57; app/models.py:81 @ 84d392ec8f0d30da
Reusable valid	README.md:82; app/main.py:233-242 @ 84d392ec8f0d30da1
Reusable valid	README.md:82; app/main.py:233-242 @ 84d392ec8f0d30da1
Reusable valid	app/models.py:94,100; migrations/versions/001_initial.sql @
Negative / not reusable	app/main.py:255-261 @ 84d392ec8f0d30da10fccdbf00ea2bc
Negative / not reusable	app/main.py:255-261 @ 84d392ec8f0d30da10fccdbf00ea2bc
Negative / not reusable	app/main.py:255-261 @ 84d392ec8f0d30da10fccdbf00ea2bc
Negative / not reusable	app/main.py:255-261 @ 84d392ec8f0d30da10fccdbf00ea2bc
Negative / not reusable	README.md:33-40; app/main.py:106-117 @ 84d392ec8f0d30
Negative / not reusable	README.md:33-40; app/main.py:106-117 @ 84d392ec8f0d30
Negative / not reusable	app/models.py:13-18; app/schemas.py:37 @ 84d392ec8f0d30
Negative / not reusable	app/models.py:13-18; app/schemas.py:37 @ 84d392ec8f0d30
Negative / not reusable	app/schemas.py:36; templates/index.html:18 @ 84d392ec8f0

T _T Dataset class	T _T Rule / source at pinned commit
Negative / not reusable	app/schemas.py:35; app/models.py:54 @ 84d392ec8f0d30da
Negative / not reusable	app/schemas.py:7,34,42-47; app/models.py:53 @ 84d392ec8f
Negative / not reusable	app/schemas.py:7,34,42-47; app/models.py:53 @ 84d392ec8f
Negative / not reusable	app/models.py:57,62; app/schemas.py:39 @ 84d392ec8f0d30c
Negative / not reusable	app/models.py:56,64-74; app/schemas.py:38 @ 84d392ec8f0
Negative / not reusable	app/models.py:56,64-74; app/schemas.py:38 @ 84d392ec8f0
Negative / not reusable	app/schemas.py:19; app/models.py:31 @ 84d392ec8f0d30da
Negative / not reusable	app/schemas.py:21; app/models.py:33 @ 84d392ec8f0d30da
Negative / not reusable	app/schemas.py:17; app/models.py:29 @ 84d392ec8f0d30da
Negative / not reusable	app/schemas.py:17; app/models.py:29 @ 84d392ec8f0d30da
Negative / not reusable	app/schemas.py:16; app/models.py:28 @ 84d392ec8f0d30da
Negative / not reusable	app/schemas.py:23; app/models.py:35 @ 84d392ec8f0d30da
Negative / not reusable	app/schemas.py:6,18,26-31 @ 84d392ec8f0d30da10fccdbf00
Negative / not reusable	app/schemas.py:22; app/models.py:34 @ 84d392ec8f0d30da
Negative / not reusable	app/models.py:7-11,36; app/schemas.py:24 @ 84d392ec8f0d
Negative / not reusable	app/models.py:7-11,36; app/schemas.py:24 @ 84d392ec8f0d
Negative / not reusable	app/models.py:29; app/main.py:182-194 @ 84d392ec8f0d30d
Negative / not reusable	app/models.py:28; app/main.py:182-194 @ 84d392ec8f0d30d
Negative / not reusable	app/main.py:245-251; templates/index.html:17 @ 84d392ec8f
Negative / not reusable	app/main.py:245-251; templates/index.html:17 @ 84d392ec8f
Negative / not reusable	app/schemas.py:10-11; app/models.py:44 @ 84d392ec8f0d30c

T _T Dataset class	T _T Rule / source at pinned commit
Negative / not reusable	app/schemas.py:13 @ 84d392ec8f0d30da10fccdbf00ea2bcfa
Negative / not reusable	app/schemas.py:13 @ 84d392ec8f0d30da10fccdbf00ea2bcfa
Negative / not reusable	app/schemas.py:9-13; app/models.py:40-48 @ 84d392ec8f0d
Negative / not reusable	app/schemas.py:9-13; app/models.py:40-48 @ 84d392ec8f0d
Negative / not reusable	app/models.py:80; app/schemas.py:51 @ 84d392ec8f0d30da
Negative / not reusable	app/models.py:80; app/schemas.py:51 @ 84d392ec8f0d30da
Negative / not reusable	app/models.py:92-93; app/schemas.py:52 @ 84d392ec8f0d30
Negative / not reusable	app/models.py:92-93; app/schemas.py:52 @ 84d392ec8f0d30
Negative / not reusable	app/schemas.py:53; README.md:82 @ 84d392ec8f0d30da10
Negative / not reusable	app/schemas.py:53; README.md:82 @ 84d392ec8f0d30da10
Negative / not reusable	app/schemas.py:57; app/models.py:81 @ 84d392ec8f0d30da
Negative / not reusable	app/models.py:20-24,82; app/schemas.py:54 @ 84d392ec8f0
Negative / not reusable	app/models.py:92,99 @ 84d392ec8f0d30da10fccdbf00ea2bc
Negative / not reusable	app/models.py:92,99 @ 84d392ec8f0d30da10fccdbf00ea2bc
Negative / not reusable	app/models.py:93,99 @ 84d392ec8f0d30da10fccdbf00ea2bc
Negative / not reusable	app/models.py:93,99 @ 84d392ec8f0d30da10fccdbf00ea2bc
Negative / not reusable	README.md:82; app/main.py:233-242 @ 84d392ec8f0d30da1
Negative / not reusable	README.md:82; app/main.py:233-242 @ 84d392ec8f0d30da1
Negative / not reusable	app/models.py:94,100; migrations/versions/001_initial.sql @
Negative / not reusable	app/models.py:94,100; migrations/versions/001_initial.sql @
Not applicable / not reusable	app/models.py created_at; app/models.py EnvironmentState (

T _T Dataset class	T _T Rule / source at pinned commit
Reusable valid	app/main.py:255-261 @ 84d392ec8f0d30da10fccdbf00ea2bc
Negative / not reusable	app/schemas.py:7,34,42-47; app/models.py:53 @ 84d392ec8f
Negative / not reusable	app/models.py:56,64-74; app/schemas.py:38 @ 84d392ec8f0
Negative / not reusable	app/schemas.py:17; app/models.py:29 @ 84d392ec8f0d30da
Negative / not reusable	app/models.py:29; app/main.py:182-194 @ 84d392ec8f0d30d
Negative / not reusable	app/schemas.py:6,26-31; app/main.py:182-194 @ 84d392ec8
Reusable valid	app/main.py:245-251; templates/index.html:17 @ 84d392ec8f
Negative / not reusable	app/schemas.py:10-11; app/models.py:44 @ 84d392ec8f0d30
Negative / not reusable	app/schemas.py:9-13; app/models.py:40-48 @ 84d392ec8f0d
Negative / not reusable	app/schemas.py:53; README.md:82 @ 84d392ec8f0d30da10
Negative / not reusable	app/schemas.py:57; app/models.py:81 @ 84d392ec8f0d30da
Negative / not reusable	app/models.py:93,99 @ 84d392ec8f0d30da10fccdbf00ea2bc
Negative / not reusable	README.md:82; app/main.py:233-242 @ 84d392ec8f0d30da1
Reusable valid	app/main.py:255-261 @ 84d392ec8f0d30da10fccdbf00ea2bc
Reusable valid	app/models.py:13-18; app/schemas.py:37 @ 84d392ec8f0d30
Reusable valid	app/schemas.py:36; templates/index.html:18 @ 84d392ec8f0
Reusable valid	app/schemas.py:19; app/models.py:31 @ 84d392ec8f0d30da
Reusable valid	app/schemas.py:21; app/models.py:33 @ 84d392ec8f0d30da
Reusable valid	app/schemas.py:16; app/models.py:28 @ 84d392ec8f0d30da
Reusable valid	app/models.py:7-11,36; app/schemas.py:24 @ 84d392ec8f0d
Reusable valid	app/schemas.py:13 @ 84d392ec8f0d30da10fccdbf00ea2bcfa

T _T	Dataset class	T _T	Rule / source at pinned commit
	Reusable valid		app/models.py:80; app/schemas.py:51 @ 84d392ec8f0d30da
	Reusable valid		app/models.py:92-93; app/schemas.py:52 @ 84d392ec8f0d30da
	Reusable valid		app/schemas.py:55; app/models.py:83 @ 84d392ec8f0d30da
	Reusable valid		app/models.py:20-24,82; app/schemas.py:54 @ 84d392ec8f0d30da

T Relationship / condition	T Execution & safety disposition	📅 Generation date
Zero delta is a safe non-changing confirmation	Executed safely in verified staging	8/5/2026
Only adjust generated product id=4	Executed safely in verified staging	8/5/2026
Human-readable run trace; not persisted	Executed safely in verified staging	8/5/2026
Dedicated public fictional staging credential only	Executed safely in verified staging	8/5/2026
Dedicated staging-only identity; never production	Executed safely in verified staging	8/5/2026
Checkbox true/false	Executed safely in verified staging	8/5/2026
office electronics home test-kit	Executed safely in verified staging	8/5/2026
Reflected text must be HTML escaped	Executed safely in verified staging	8/5/2026
Run-tagged fictional order-management merchandis	Executed safely in verified staging	8/5/2026
^[A-Z0-9][A-Z0-9-]{2,23}\$; unique	Executed safely in verified staging	8/5/2026
Available stock; dependent order quantity cannot exc	Executed safely in verified staging	8/5/2026
ORM @validates requires >0; <=999999.99; two decir	Executed safely in verified staging	8/5/2026
Fictional staging shipping address	Executed safely in verified staging	8/5/2026
Optional unit/suite	Executed safely in verified staging	8/5/2026
Fictional location	Executed safely in verified staging	8/5/2026
IANA-reserved example.com; unique	Executed safely in verified staging	8/5/2026
Run-tagged realistic fictional customer	Executed safely in verified staging	8/5/2026
Run tag; no external actions	Executed safely in verified staging	8/5/2026
^\+?[0-9 ()-]{7,24}\$; fictional 555-01xx	Executed safely in verified staging	8/5/2026
Fictional postal code	Executed safely in verified staging	8/5/2026
active inactive suspended	Executed safely in verified staging	8/5/2026

T┐ Relationship / condition	T┐ Execution & safety disposition	📅 Generation date
Preserve the generated customer's own unique email	Executed safely in verified staging	8/5/2026
Only update this run's generated customer	Executed safely in verified staging	8/5/2026
Run-tagged existing customer only	Executed safely in verified staging	8/5/2026
Same phone format rule as Customer create	Executed safely in verified staging	8/5/2026
First failure must pass exact replay before being treated as failure	Executed safely in verified staging	8/5/2026
Inline tab shares the dashboard URL	Executed safely in verified staging	8/5/2026
Run identifier provides record traceability	Executed safely in verified staging	8/5/2026
RFC email; reserved example.com domain; unique	Executed safely in verified staging	8/5/2026
qa_viewer qa_editor; default qa_viewer	Executed safely in verified staging	8/5/2026
^[a-zA-Z][a-zA-Z0-9_-]+\$; unique	Executed safely in verified staging	8/5/2026
Existing customers.id=4; own run-tagged customer	Executed safely in verified staging	8/5/2026
Carries run tag for orders and inherited order-item traceability	Executed safely in verified staging	8/5/2026
Existing products.id=4; available stock=8	Executed safely in verified staging	8/5/2026
Must also be <= referenced product stock	Executed safely in verified staging	8/5/2026
Never blank for reusable records	Executed safely in verified staging	8/5/2026
standard express pickup	Executed safely in verified staging	8/5/2026
draft pending paid shipped cancelled; draft avoids existing	Executed safely in verified staging	8/5/2026
Existing run-tagged orders.id=2	Executed safely in verified staging	8/5/2026
Existing run-tagged orders.id=2	Executed; no orphan	8/5/2026
Existing products.id=2; pair (order_id,product_id) unique	Executed safely in verified staging	8/5/2026
Product 2 stock=4; preserve FK and unique pair	Executed safely in verified staging	8/5/2026

T Relationship / condition	T Execution & safety disposition	📅 Generation date
>=0; capture product price	Executed safely in verified staging	8/5/2026
Zero delta is a safe non-changing confirmation	Generated; not executed to preserve stock	8/5/2026
Human-readable run trace; not persisted	Generated; relationship-safe positive variant	8/5/2026
Checkbox true/false	Generated; do not use inactive product in an or	8/5/2026
Reflected text must be HTML escaped	Generated; relationship-safe positive variant	8/5/2026
^[A-Z0-9][A-Z0-9-]{2,23}\$; unique	Generated; use a unique SKU when executed	8/5/2026
^[A-Z0-9][A-Z0-9-]{2,23}\$; unique	Generated; ensure uniqueness	8/5/2026
Available stock; dependent order quantity cannot exc	Generated; never create an order against unav	8/5/2026
Available stock; dependent order quantity cannot exc	Generated; no fulfillment actions	8/5/2026
ORM @validates requires >0; <=999999.99; two decir	Generated; no external payments	8/5/2026
ORM @validates requires >0; <=999999.99; two decir	Generated; staging only; no billing integration	8/5/2026
Fictional staging shipping address	Generated; retain fictional domain context	8/5/2026
Optional unit/suite	Generated; relationship-safe positive variant	8/5/2026
Fictional location	Generated	8/5/2026
Run-tagged realistic fictional customer	Generated only; run tag must be in notes	8/5/2026
Run tag; no external actions	Generated; relationship-safe positive variant	8/5/2026
Run tag; no external actions	Executed safely	8/5/2026
^\+?[0-9 ()-]{7,24}\$; fictional 555-01xx	Generated; relationship-safe positive variant	8/5/2026
^\+?[0-9 ()-]{7,24}\$; fictional 555-01xx	Generated; no relationship impact	8/5/2026
Fictional postal code	Generated	8/5/2026
Run-tagged existing customer only	Generated; relationship-safe positive variant	8/5/2026

T┐ Relationship / condition	T┐ Execution & safety disposition	📅 Generation date
Same phone format rule as Customer create	Generated; relationship-safe positive variant	8/5/2026
Run identifier provides record traceability	Generated; relationship-safe positive variant	8/5/2026
^[a-zA-Z][a-zA-Z0-9_-]+\$; unique	Executed safely; run tag in display_name	8/5/2026
^[a-zA-Z][a-zA-Z0-9_-]+\$; unique	Executed safely; run tag in display_name	8/5/2026
Carries run tag for orders and inherited order-item tra	Generated; relationship-safe positive variant	8/5/2026
Carries run tag for orders and inherited order-item tra	Generated; parent tag traces child items	8/5/2026
Must also be <= referenced product stock	Use nonempty shipping in reusable dataset	8/5/2026
Use products.id=3; current stock 100	Generated; customer and product must already	8/5/2026
Never blank for reusable records	Generated; no external fulfillment	8/5/2026
Product 2 stock=4; preserve FK and unique pair	Executed; product stock 4	8/5/2026
Product 2 stock=4; preserve FK and unique pair	Generated; use a currently unused (order_id,pr	8/5/2026
>=0; capture product price	Generated; use valid related order/product pair	8/5/2026
Zero delta is a safe non-changing confirmation	Generated only; no unsafe write	8/5/2026
Zero delta is a safe non-changing confirmation	Executed safely; no inventory update	8/5/2026
Only adjust generated product id=4	Generated only; no unsafe write	8/5/2026
Only adjust generated product id=4	Manual review; browser select offers existing I	8/5/2026
Dedicated public fictional staging credential only	Generated only; no unsafe write	8/5/2026
Dedicated staging-only identity; never production	Generated only; no unsafe write	8/5/2026
office electronics home test-kit	Generated only; no unsafe write	8/5/2026
office electronics home test-kit	Manual review; no UI tampering	8/5/2026
Reflected text must be HTML escaped	Executed inert markup only; GAP-04 / TCW-91	8/5/2026

T Relationship / condition	T Execution & safety disposition	📅 Generation date
Run-tagged fictional order-management merchandis	Generated only; no unsafe write	8/5/2026
^[A-Z0-9][A-Z0-9-]{2,23}\$; unique	Generated only; no unsafe write	8/5/2026
^[A-Z0-9][A-Z0-9-]{2,23}\$; unique	Executed; no record created	8/5/2026
Available stock; dependent order quantity cannot exc	Generated; no invalid inventory record	8/5/2026
ORM @validates requires >0; <=999999.99; two decir	Generated only; no unsafe write	8/5/2026
ORM @validates requires >0; <=999999.99; two decir	Not submitted; covered by repeated -3.25 prob	8/5/2026
Fictional staging shipping address	Generated only; no unsafe write	8/5/2026
Fictional location	Generated only; no unsafe write	8/5/2026
IANA-reserved example.com; unique	Generated only; no unsafe write	8/5/2026
IANA-reserved example.com; unique	Generated; no external email or record	8/5/2026
Run-tagged realistic fictional customer	Generated only; no unsafe write	8/5/2026
Run tag; no external actions	Generated only; never forced past browser con	8/5/2026
^\+?[0-9 ()-]{7,24}\$; fictional 555-01xx	Executed safely; GAP-02 / TCW-90	8/5/2026
Fictional postal code	Generated only; no unsafe write	8/5/2026
active inactive suspended	Generated only; no unsafe write	8/5/2026
active inactive suspended	Manual review; never force an unsupported op	8/5/2026
Preserve the generated customer's own unique emai	Generated only; no unsafe write	8/5/2026
Only update this run's generated customer	Generated only; no unsafe write	8/5/2026
First failure must pass exact replay before being trea	Generated only; no unsafe write	8/5/2026
Inline tab shares the dashboard URL	Generated only; no unsafe write	8/5/2026
RFC email; reserved example.com domain; unique	Generated only; no unsafe write	8/5/2026

T┐ Relationship / condition	T┐ Execution & safety disposition	📅 Generation date
qa_viewer qa_editor; default qa_viewer	Generated only; no unsafe write	8/5/2026
qa_viewer qa_editor; default qa_viewer	Manual review; do not tamper with DOM/selec	8/5/2026
^[a-zA-Z][a-zA-Z0-9_-]+\$; unique	Generated only; no unsafe write	8/5/2026
^[a-zA-Z][a-zA-Z0-9_-]+\$; unique	Generated only; no record written	8/5/2026
Existing customers.id=4; own run-tagged customer	Generated only; no unsafe write	8/5/2026
Existing customers.id=4; own run-tagged customer	Manual review; do not force invalid option	8/5/2026
Existing products.id=4; available stock=8	Generated only; no unsafe write	8/5/2026
Existing products.id=4; available stock=8	Manual review; no invalid select tampering	8/5/2026
Must also be <= referenced product stock	Generated only; no unsafe write	8/5/2026
Must also be <= referenced product stock	Executed; no order created	8/5/2026
Never blank for reusable records	Executed twice; no orphan; GAP-01	8/5/2026
draft pending paid shipped cancelled; draft avoids ex	Manual review; no unsupported actions	8/5/2026
Existing run-tagged orders.id=2	Generated only; no unsafe write	8/5/2026
Existing run-tagged orders.id=2	BLOCKED / manual review: never submit orphæ	8/5/2026
Existing products.id=2; pair (order_id,product_id) uni	Generated only; no unsafe write	8/5/2026
Existing products.id=2; pair (order_id,product_id) uni	BLOCKED / manual review: no orphan records;	8/5/2026
Product 2 stock=4; preserve FK and unique pair	Generated only; no unsafe write	8/5/2026
products.id=2 stock=4	Covered by tagged Order quantity replays; no c	8/5/2026
>=0; capture product price	Generated only; no unsafe write	8/5/2026
>=0; capture product price	Generated only; no record created	8/5/2026
System-managed timestamps only	Not executed; manual review if an editable dat	8/5/2026

T Relationship / condition	T Execution & safety disposition	📅 Generation date
Zero delta is a safe non-changing confirmation	Executed; no domain record created	8/5/2026
^[A-Z0-9][A-Z0-9-]{2,23}\$; unique	Executed safely; no extra record	8/5/2026
ORM @validates requires >0; <=999999.99; two decim	Executed twice with distinct required unique S	8/5/2026
IANA-reserved example.com; unique	Executed; no record created	8/5/2026
Preserve the generated customer's own unique email	Executed twice; no email change persisted; GA	8/5/2026
Same phone format rule as Customer create	Executed twice on this run's customer; GAP-02	8/5/2026
First failure must pass exact replay before being trea	Executed twice; no records created	8/5/2026
RFC email; reserved example.com domain; unique	Executed twice without persistence; GAP-07 /	8/5/2026
^[a-zA-Z][a-zA-Z0-9_-]+\$; unique	Executed; no record created	8/5/2026
Existing products.id=2; available stock 4	Executed twice with valid references; GAP-05 /	8/5/2026
Never blank for reusable records	Executed twice; all FK relationships preserved;	8/5/2026
Existing products.id=2; pair (order_id,product_id) uni	Generated; avoid replaying an existing pair	8/5/2026
Product 2 stock=4; preserve FK and unique pair	Executed twice with existing order/product IDs	8/5/2026
Human-readable run trace; not persisted	Generated; reason is not separately persisted	8/5/2026
office electronics home test-kit	Safe positive variants	8/5/2026
Reflected text must be HTML escaped	Generated; output must be escaped	8/5/2026
Fictional staging shipping address	Executed safely	8/5/2026
Fictional location	Executed safely	8/5/2026
Run-tagged realistic fictional customer	Executed safely; reserved example.com email	8/5/2026
active inactive suspended	Generated positive variant	8/5/2026
qa_viewer qa_editor; default qa_viewer	Executed safely	8/5/2026

T↑ Relationship / condition	T↑ Execution & safety disposition	📅 Generation date
Existing customers.id=4; own run-tagged customer	Executed; no orphan	8/5/2026
Existing products.id=2; available stock=4	Executed with existing product	8/5/2026
standard express pickup	Generated only	8/5/2026
draft pending paid shipped cancelled; draft avoids ex	Executed; all orders stayed draft	8/5/2026

 Run date	T _T Run identifier	T _T Project / repository
8/5/2026 14:20:52	QA-sample-qa-test-project-20260805T14205	sample-qa-test-project

T_T Requested branch	T_T Exact commit SHA	T_T Run status
agent/qa-validation-fixture	84d392ec8f0d30da10fccdbf00ea2bcfae3922c5	COMPLETED WITH FINDINGS

T T Staging verification

T T Workflow stage stopped

Verified repo-defined localhost staging; API environment=staging, prod Completed; Teams notification sent to Testing

T_T Failure / blocker reason

T_T Forms discovered

9

Forms tested	T _T Forms skipped due to cap	Records created in staging	T _T Validation gaps confirmed	Jira issues created
9	None (0); all forms below 25-form cap.	24	6	6

T_T Google Sheet	T_T QA report	T_T Jira issues
https://docs.google.com/spreadsheets/d/1X	https://docs.google.com/document/d/1X	TCW-89: https://workflow-testing- TCW-90: https://workflow-testing- TCW-91: https://workflow-testing- TCW-92: https://workflow-testing- TCW-93: https://workflow-testing- TCW-94: https://workflow-testing-

T_r Notes

Critical 1; High 3; Medium 1; Low 1. GAP-03 blocked/manual review